Comparative Evaluation of Continual Learning Methods on Rotated MNIST

1. Dataset Selection: Why Rotated MNIST?

The Rotated MNIST dataset presents a controlled domain shift via rotation angles, providing a meaningful continual learning scenario. Unlike standard MNIST, it allows for future task generalization (FWT) evaluation, due to its inherent gradual task transformations. It is lightweight, visually intuitive, and widely used in lifelong learning benchmarks.

2. Method Categories and Motivation

The selected methods span three primary categories:

- Regularization-based:
  - EWC: Adds Fisher-based penalty to prevent forgetting important parameters.
  - MAS: Uses gradient-based importance measures for parameter constraints.

- Replay-based:
  - Replay: Stores and reuses previous task data.
  - DER: Combines replay with knowledge distillation.

- Optimization-based:
  - GEM: Projects gradients to avoid interference with previous tasks.
  - OGD: Orthogonalizes gradients via geometry preservation.

3. Quantitative Results (Rotated MNIST)

Method	Forgetting	BWT	FWT
Vanilla	22.80%	-22.80%	61.26%
EWC	23.58%	-23.58%	60.25%
MAS	10.58%	-10.58%	51.18%
Replay	7.45%	-7.45%	58.84%
DER	26.72%	-26.72%	61.06%

OGD | 19.98% | -21.48% | 73.18%

GEM | 22.32% | -24.44% | 73.47%

4. Analysis and Comparison

Replay-based methods (Replay, DER) show robust resistance to forgetting due to explicit memory usage, but DER's gradient noise can hurt generalization. GEM performs competitively but is memory and computation heavy.

OGD strikes a balance by avoiding gradient conflict geometrically, yielding high FWT. Regularization methods like MAS significantly reduce forgetting, though at the cost of flexibility.

5. Conclusion and Future Work

Rotated MNIST has demonstrated itself as an informative benchmark for evaluating multiple continual learning mechanisms under task-incremental scenarios. Results indicate the strong trade-off between forgetting mitigation and forward transfer.

Next steps include expanding experiments to semantic CIFAR-100 and TinyImageNet splits to evaluate scalability and domain generalization.